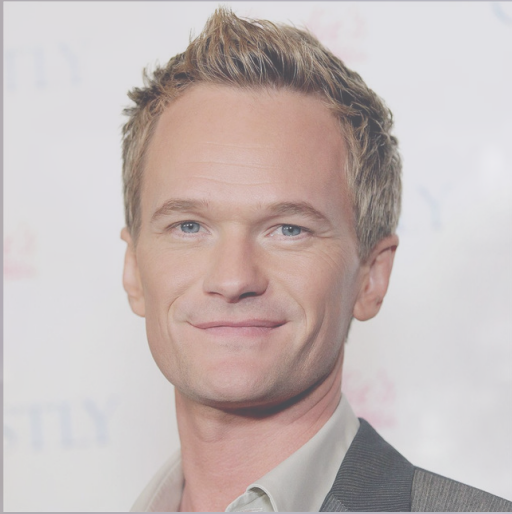


What are the foundations of academic success?



Doogie Howser, M.D., sitcom from 1989. Paraphrase from Wikipedia:
'Possessing a genius intellect and a photographic memory, Howser earned a perfect score on the SAT at the age of 6, completed high school in nine weeks at the age of 9, graduated from Princeton University at age 10, and finished medical school four years later. At age 14, Howser was the youngest licensed doctor in the country. He can't buy beer, but he can prescribe drugs.'

Midterm: 9:35-10:25am on Thursday October 9

- Exam will be during our usual class time. Arrive a few mins early if possible.
- Exam will be hand-written, not typed.
- All content from lectures (including this one), textbook, precept, & readings is fair game. Focus on what the textbook is highlighting, not on those endless tiny details.
- Do not memorize names of researchers or specific kid ages.
- Format: 20 multiple choice, 12 fill-in-the-blank, 1 short-answer question
- Review session during precept: You'll review lecture/precept/reading material in your precept today/tomorrow. This is intended to remind you of some topics we've covered.
- Practice questions: Also, check out the practice MCQs for each chapter in the textbook.
- You will enter your Student ID **number** on the exam. Please bring your ID card.

LIFE

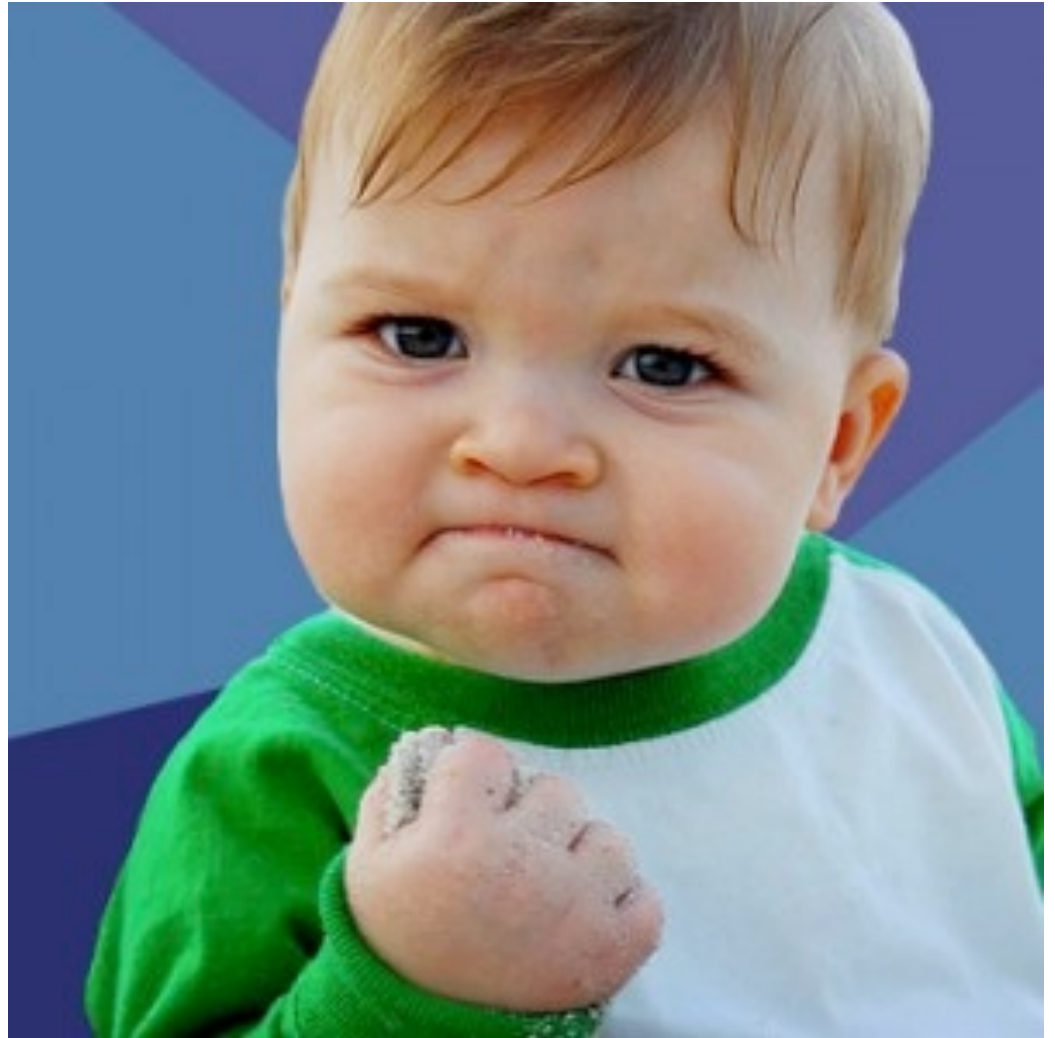
Babies are **SMARTER** than you Think

They can
Add
before they can
Count.
They can
Understand
a hundred words
before they can
Speak.
And, at three months,
their powers of
Memory
are far greater
than we ever
imagined.



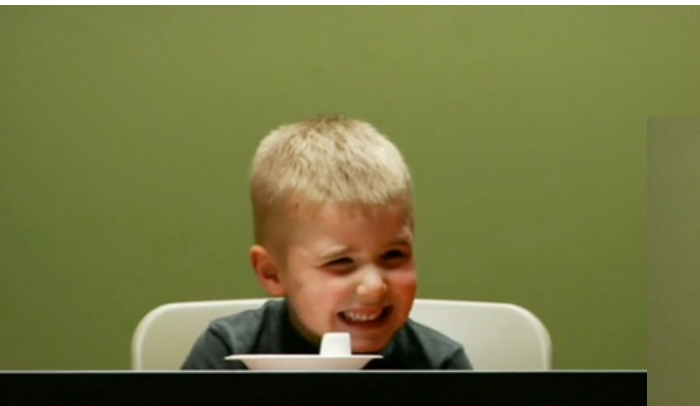
Today

- Delay of gratification
- Math and Reading
- IQ
- Socioeconomic status & the achievement gap



Delay of gratification

Oh, The Temptation
THE MARSHMALLOW TEST



Delay of gratification



But children's willingness to wait depends on how reliable the adult is!

(Kidd, Palmeri, & Aslin, 2013)

Delay of gratification

sciencealert

Trending



Sepia officinalis. (Schafer & Hill/The Image Bank/Getty Images)

NATURE

A Cephalopod Has Passed a Cognitive Test Designed For Human Children

MICHELLE STARR 3 MARCH 2021

Delay of gratification

- It's (weakly) predictive of some long-term outcomes, including social, cognitive, and emotional functioning in adulthood



- More attentive
- Better concentration
- Greater self-control
- Better frustration tolerance
- Higher SATs
- Perceived as more interpersonally competent by parents and peers
- Less likely to use drugs

(Michaelson & Munakata, 2020; Shoda et al., 1990)

But wait...

- Later studies found limited/no support for the idea that delaying gratification leads to better outcomes.
- Instead, the ability to wait for a second marshmallow is shaped a lot by a child's social and economic background.
- This background — not the moment-to-moment ability to delay gratification — may be what's behind kids' long-term success.
- **Self-control alone can't overcome social and economic disadvantage.**

Today

- Delay of gratification
- **Math and Reading**
- IQ
- Socioeconomic status & the achievement gap



Math

- Number estimation in infancy predicts math ability at age 4
- Principles of counting
 - One-one: assign 1 number per object
 - Stable-order: numbers should be counted in the same order
 - Cardinality: last number = number in set
 - Order irrelevance: objects can be counted in any order
 - Abstraction: the other principles apply to any set of objects/events



Math

- Strategies for simple arithmetic

Counting from one

Counting on

Retrieval

Breaking down to easier problems

Decomposing



Math

- Math disability
 - ~6% of children
 - Poor memorization and retrieval
 - Inaccurate strategies
 - Slow development of strategies
 - Why?
 - Limited early exposure to number
 - Limited conceptual understanding
 - Limited working memory capacity

Reading

☰ SECTIONS



The New York Times



U.S.

💬 540 COMMENTS

Pediatrics Group to Recommend Reading Aloud to Children From Birth

By MOTOKO RICH JUNE 24, 2014

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🐦 Tweet

📁 Save

➦ More

In between dispensing advice on breast-feeding and immunizations, doctors will tell parents to read aloud to their infants from birth, under a new policy that the American Academy of Pediatrics will announce on Tuesday.

With the increased recognition that an important part of brain development occurs within the first three years of a



Dr. Leora Mogilner, a pediatrician at Mount Sinai Hospital, gave a book to Kaylee Smith, 9 months,

Reading

- The single best behavioral predictor of reading proficiency is early phonological awareness: the ability to attend to, reflect on, and manipulate the sounds of speech.
- Children with reading difficulties are consistently more impaired in phonological awareness than in any other single ability.
- Recognizing rhymes
- Identifying the beginning sounds in words
- Blending sounds together
- Segmenting a word into its individual sounds/syllables
- Deleting a sound



Reading

- Benchmarks of phonological awareness

<i>Age/Grade</i>	<i>Skill or ability</i>
Preschool	Some rhyming ability (e.g., match rhymes) Some beginning sound ability (e.g., beginning sound sorts)
Early kindergarten	Segment words into syllables Judge and match rhyming words Generate rhyming words
Middle kindergarten	Match words with same beginning sounds Match words with same final sounds Segment initial sounds and final sounds
Late kindergarten	Segment and blend two and three sound words that are consonant-vowel (e.g., go), vowel-consonant (e.g., up), consonant-vowel-consonant (e.g., cat, moon)
Early first grade	Segment and blend sounds in words with consonant blends (e.g., skate, jump)

Reading

- Breaking words into sounds



Reading

- Breaking words into syllables



Reading

Infancy - kindergarten

- Identify letters, write name, read a few words

1st - 3rd grade

- Translate letters into sounds, blend sounds into words
- Identify individual words more quickly (with fairly high demands)

4th - 8th grade

- “reading to learn” rather than “learning to read”

High school, college

- Reading more sophisticated range of subjects; history, politics, philosophy, fiction, social media 😊

Reading

- Alphabetic principle: understanding that sounds in words can be represented with symbols



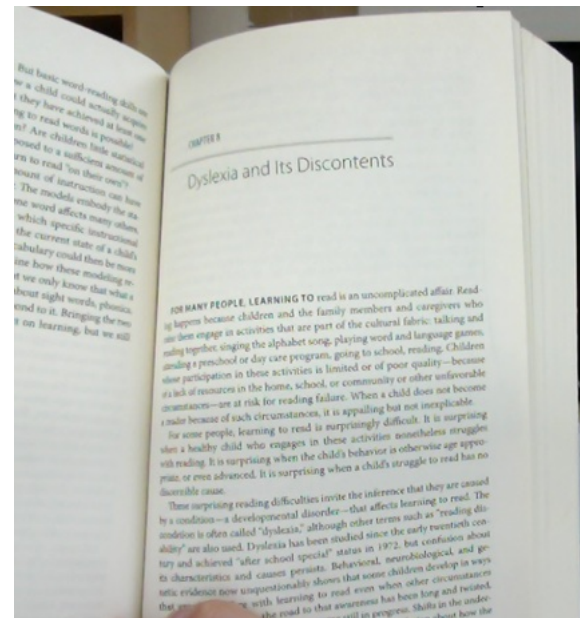
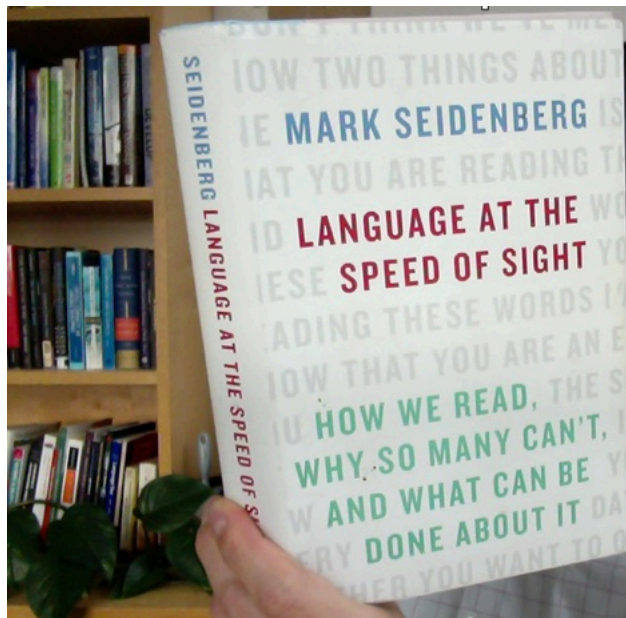
Reading

- Reading wars
 - Whole-word / meaning-based approach: skilled readers don't sound out to identify words, so kids shouldn't learn to.
 - Phonics: use sounds, letters, and words kids already know; link together previous knowledge for support.
 - *didaskaleinophobia*
 - Best recipe: more weight on phonics, but still some weight on whole-word. Polarization between these isn't the right approach.
 - For evidence of how troubled this issue is, listen to this podcast: <https://revealnews.org/podcast/how-teaching-kids-to-read-went-so-wrong-update-2023/>

Reading

- **Dyslexia**

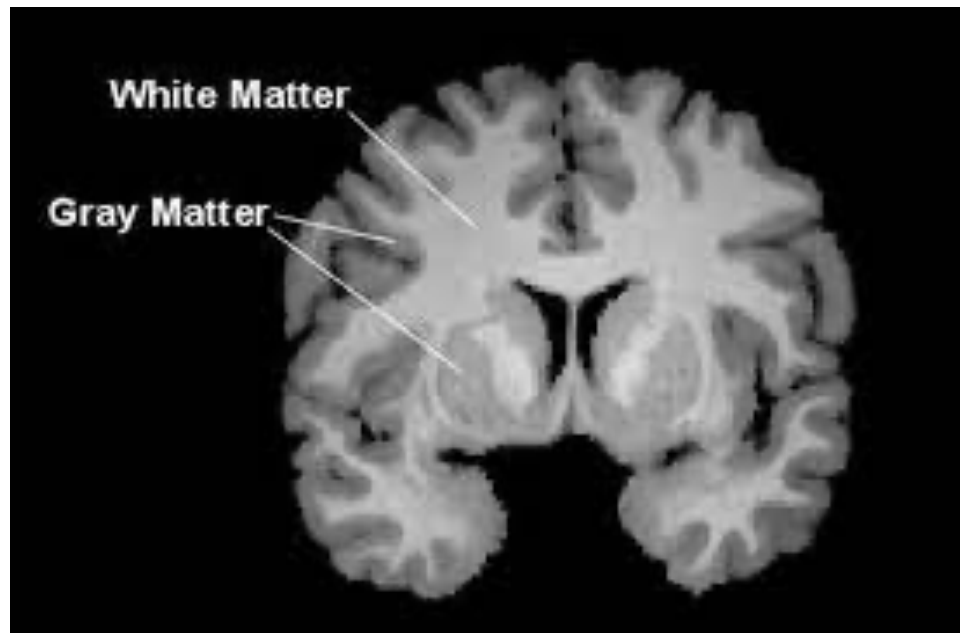
- Phonological dyslexia: difficulty sounding out/decoding, phonological awareness
 - Hard to read nonsense words (e.g., *naichovabe*)
- Surface dyslexia: difficulty with visually based retrieval
 - Hard to read exception words & words with unusual sound-letter combos (e.g., pint, yacht, caught)



Reading

- The best neural predictor of reading growth between kindergarten and 3rd grade (controlling for genetic risk, environmental factors, early language abilities, and overall cognitive abilities) is:

white matter growth in the left temporoparietal junction (TPJ)



IQ

Wechsler Intelligence Scale for Children (WISC)

Information:	On what continent is Argentina?
Arithmetic:	If 4 toys cost 6 dollars, how much do 7 toys cost?
Vocabulary:	What does “debilitating” mean?
Comprehension:	Why are streets usually numbered in order?
Picture completion:	Indicate the missing part from an incomplete picture
Block design:	Use blocks to replicate a two-color design
Object assembly:	Assemble puzzles depicting common objects
Coding:	Using a key, match symbols with shapes or numbers
Picture arrangement:	Reorder a set of scrambled pictures to tell a story
Similarities:	In what way are dogs and rabbits alike?

IQ

Leiter-R Brief IQ

4 subtests:

Locate objects in increasingly complex scenes

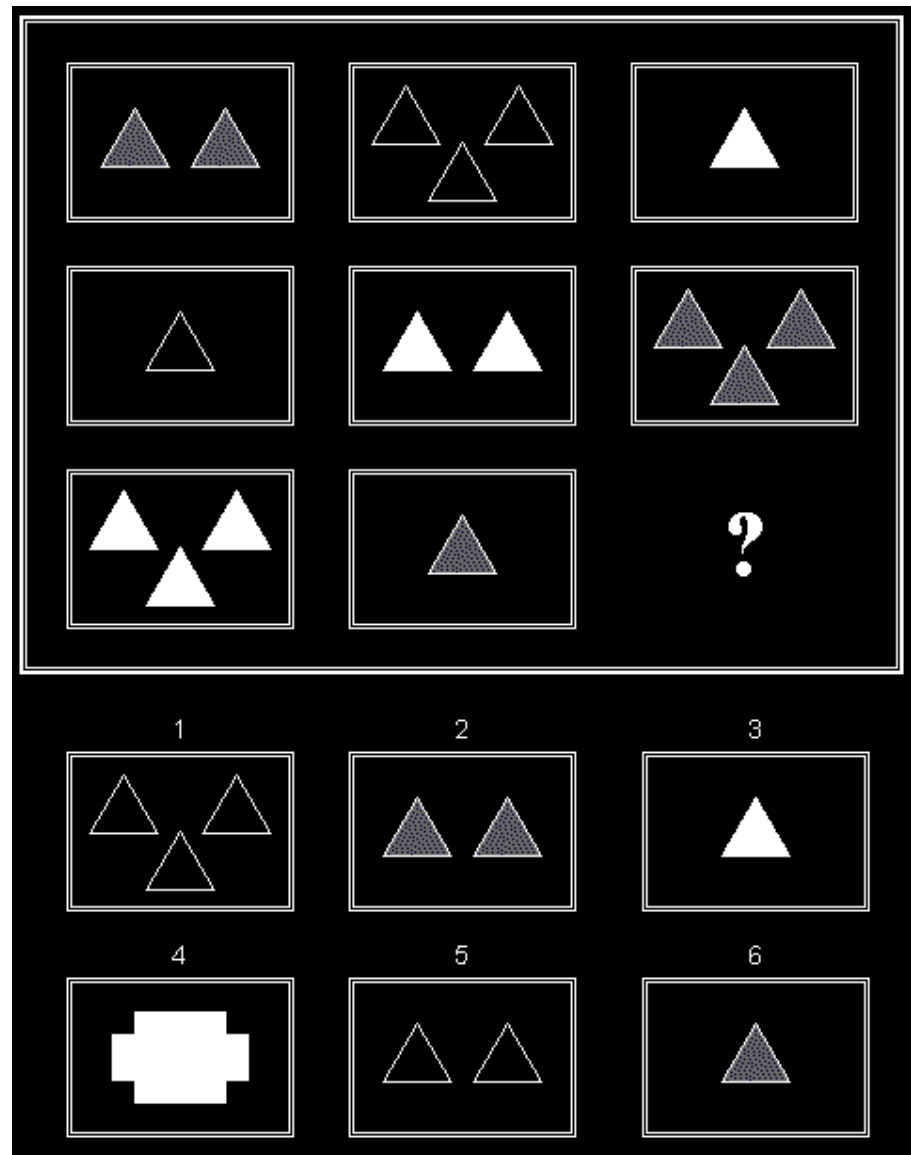
Visually integrate object parts in order to identify a whole object

Arrange a series of related pictures into a logical order

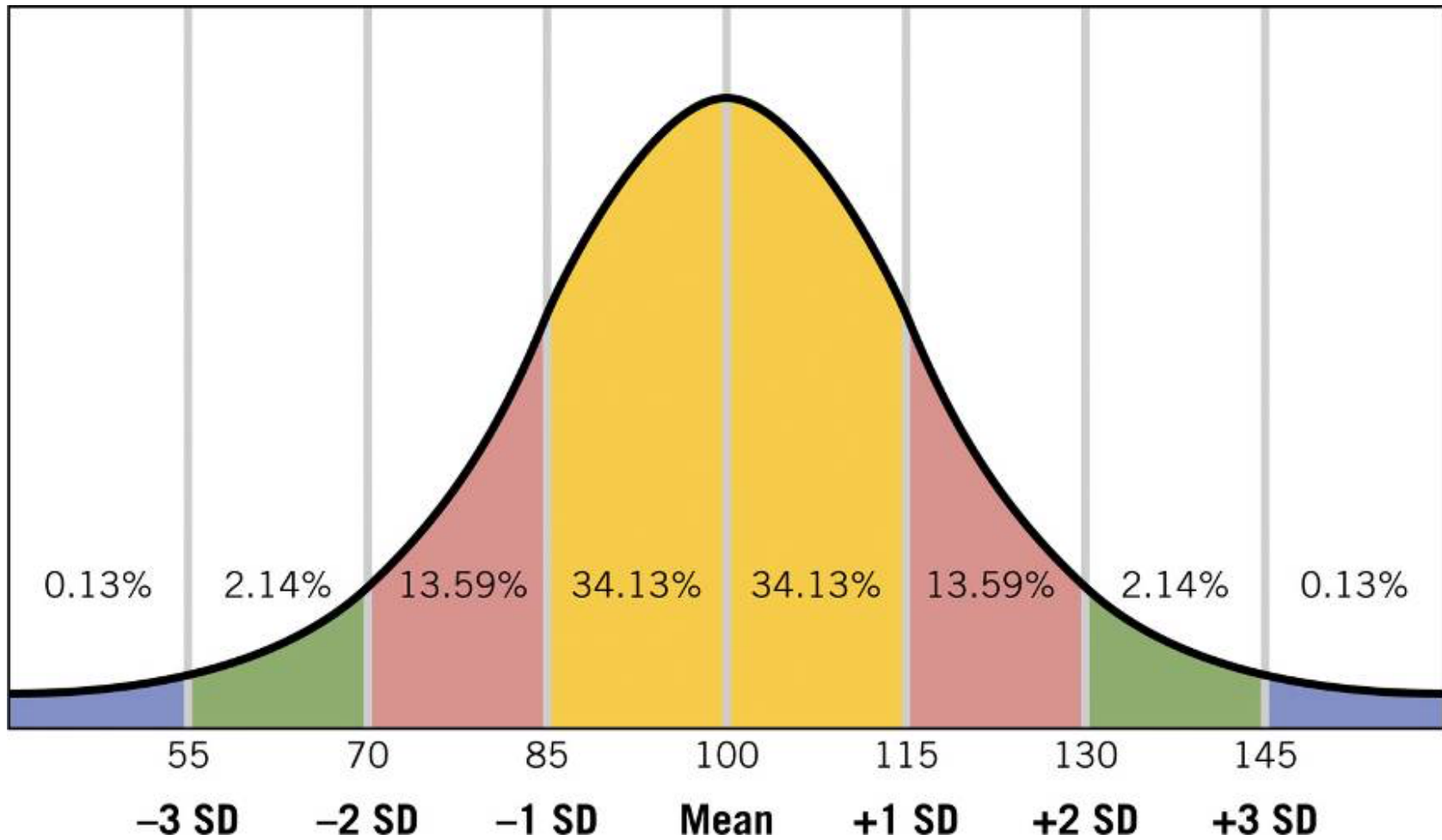
Recognize increasingly complex patterns and complete missing parts



Raven's progressive matrices



IQ



IQ

- IQ predicts academic, economic, and occupational success
- **But many other factors predict these too**: motivation, conscientiousness, curiosity, creativity, physical health, mental health, social skills, delay of gratification, practical/social intelligence...



By age 5 years, it is possible to predict,
with startling accuracy,
who will complete high school and who will not.

James Heckman
Schools, Skills, and Synapses

IQ

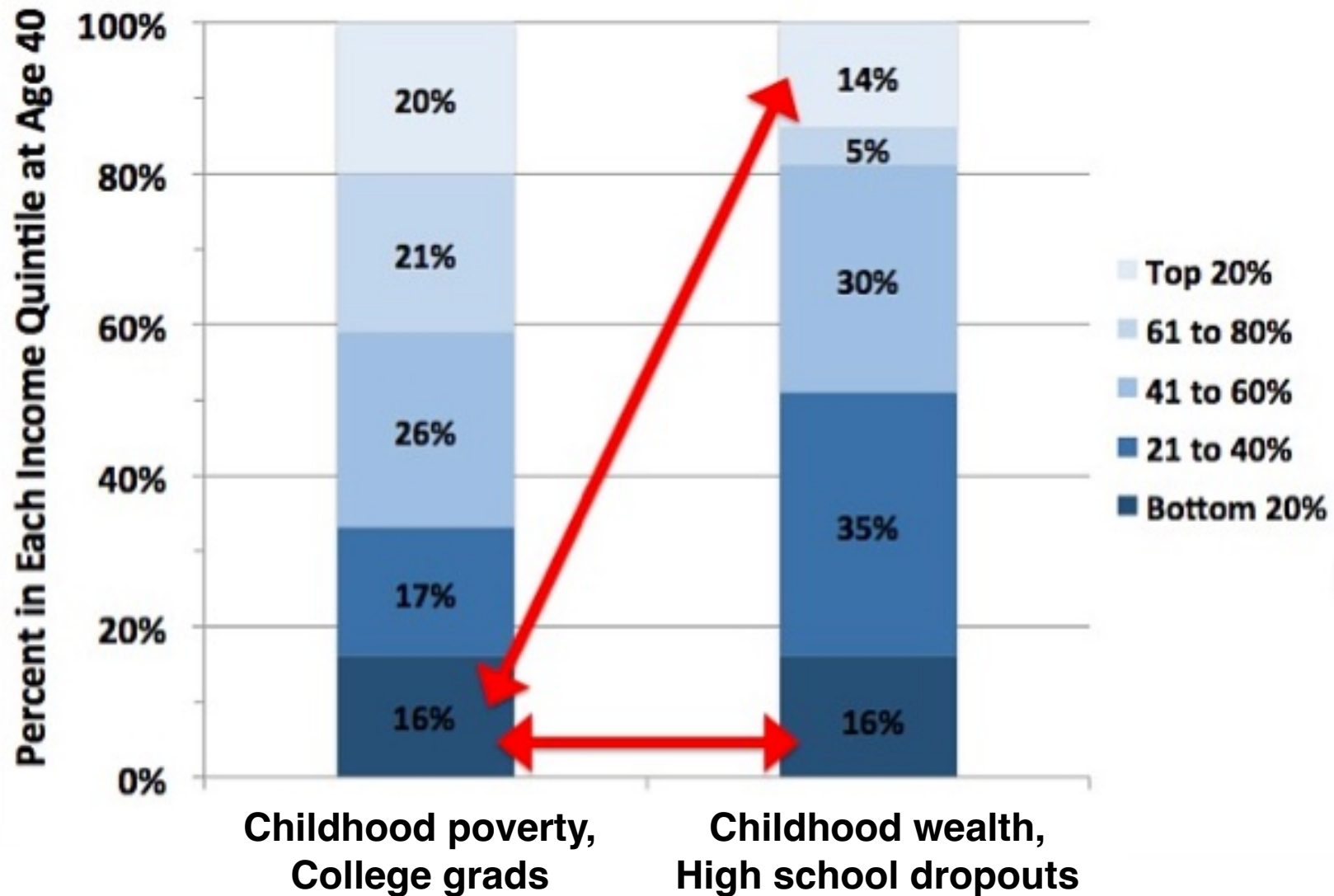
Average Familial IQ Correlations (R)

Relationship	Average R
Reared-together biological relatives	
MZ twins	0.86
DZ twins	0.60
Siblings	0.47
Parent-offspring	0.42
Half-siblings	0.35
Cousins	0.15
Reared-apart biological relatives	
MZ twins	0.72
Siblings	0.24
Parent-offspring	0.24
Reared-together nonbiological relatives	
Siblings	0.32
Parent-offspring	0.24

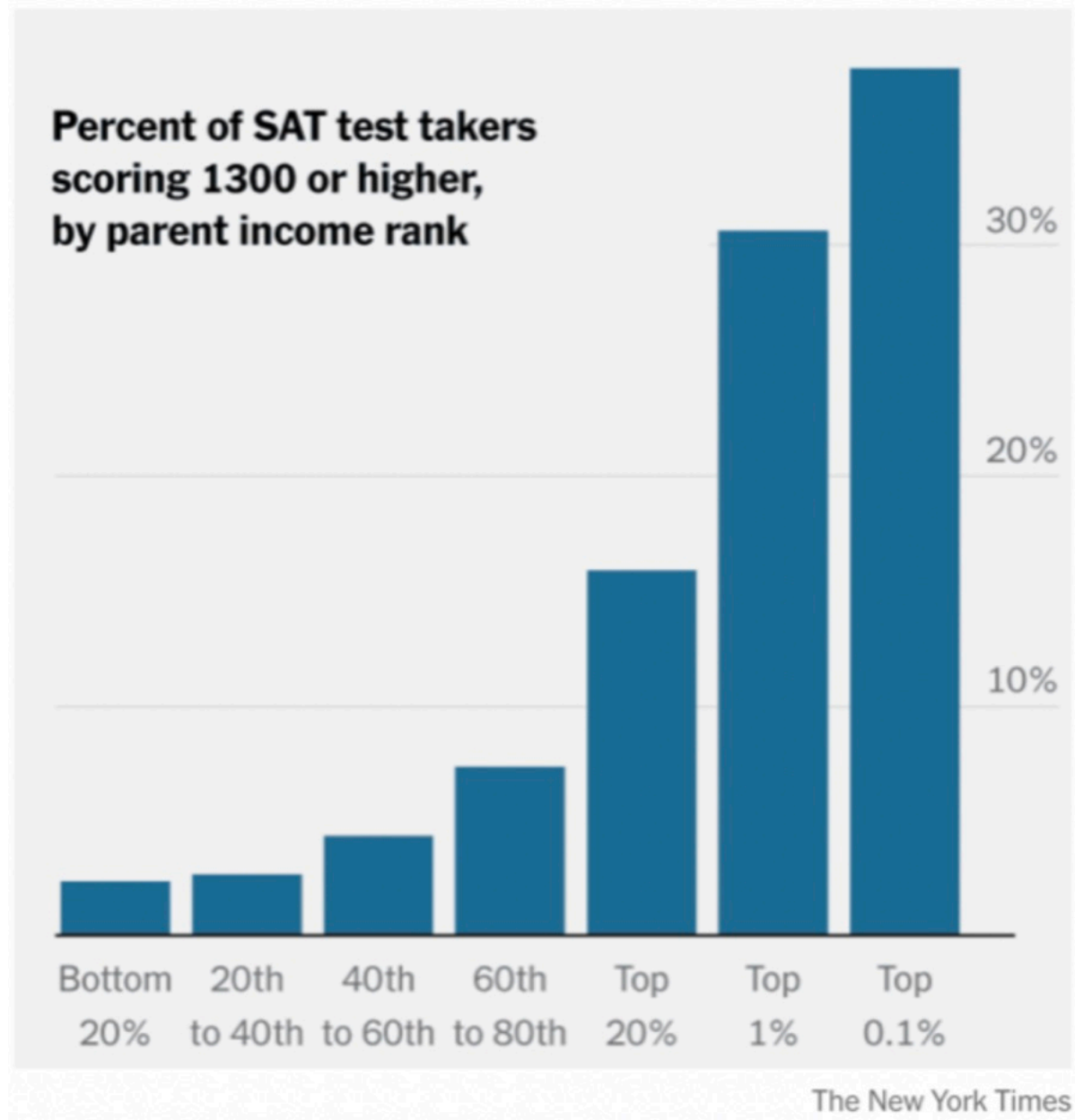
IQ



Poor kids who do everything right
don't do better than rich kids who do everything wrong



SES & SAT



The social class achievement gap in college

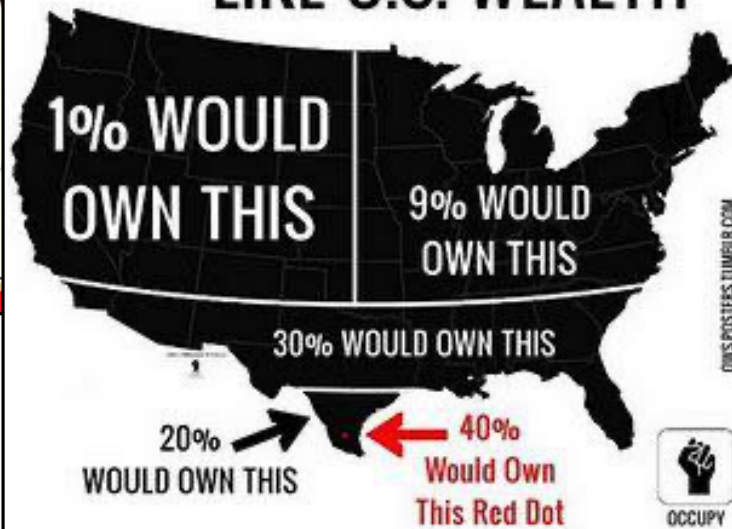
CAPITAL

in the Twenty-First Century

THOMAS
PIKETT

TRANSLATED BY ARTHUR GOLDHAUS

IF U.S. LAND WERE DIVIDED
LIKE U.S. WEALTH



**WHITHER
OPPORTUNITY?**

The New York Times Magazine



Who Gets to Graduate?

Here's a basic truth about American colleges: Rich kids complete their degrees; poor and working-class kids usually don't. A new experiment at the University of Texas is trying to change that equation. BY PAUL TOUGH

The social class achievement gap in college

- First-generation students:
 - have higher drop-out rates
 - get lower grades
 - interact less with professors
 - make fewer friends
- Obstacles to college success:
 - financial: fewer financial and material resources
 - academic: less preparation from home and high school
 - psychological: lack of fit, know-how, belonging, inclusion

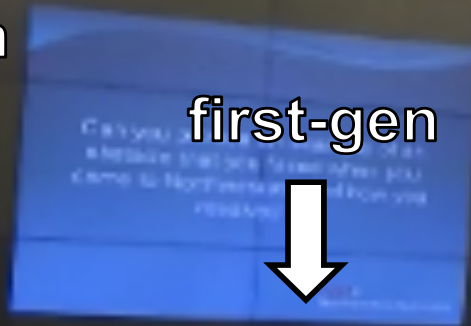
first-gen



first-gen



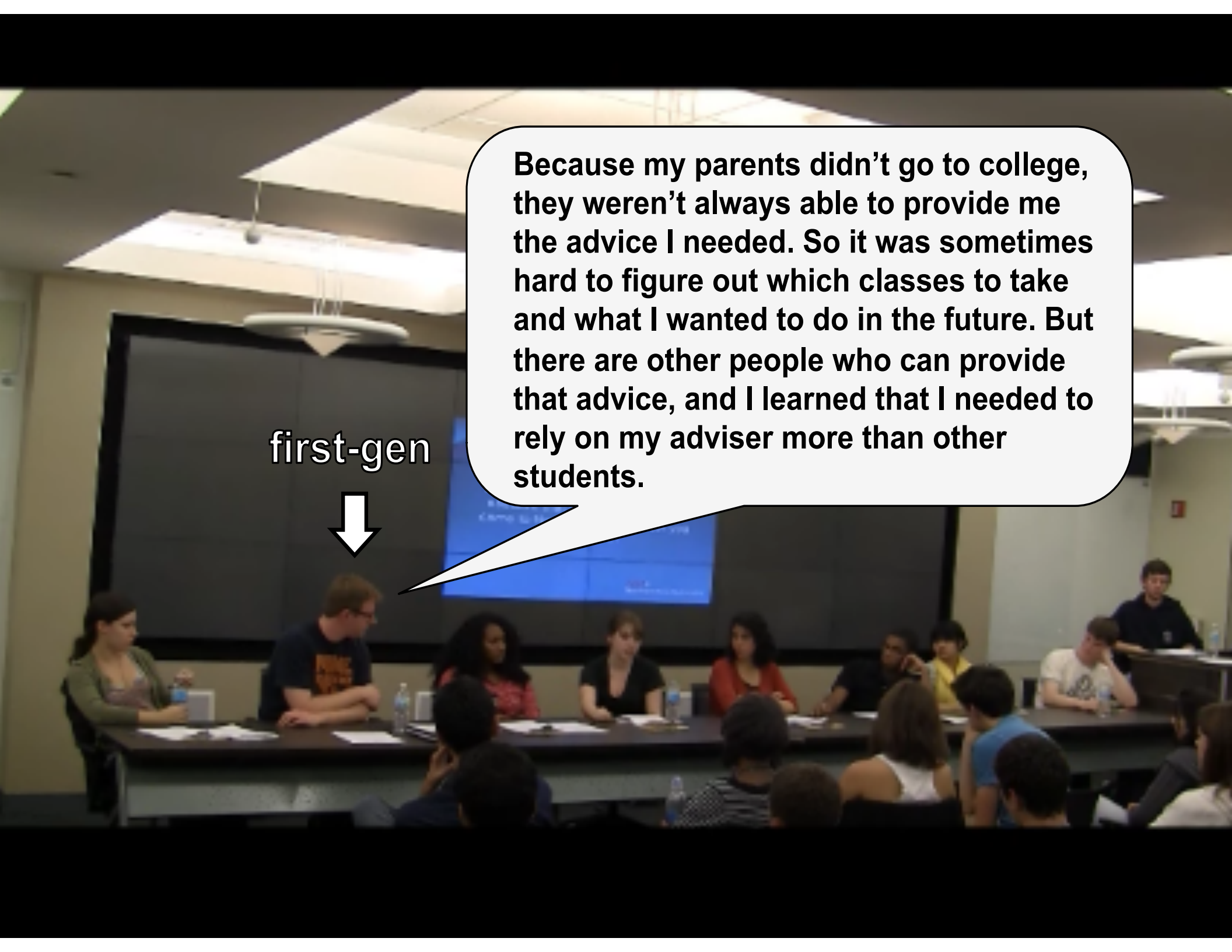
first-gen



first-gen

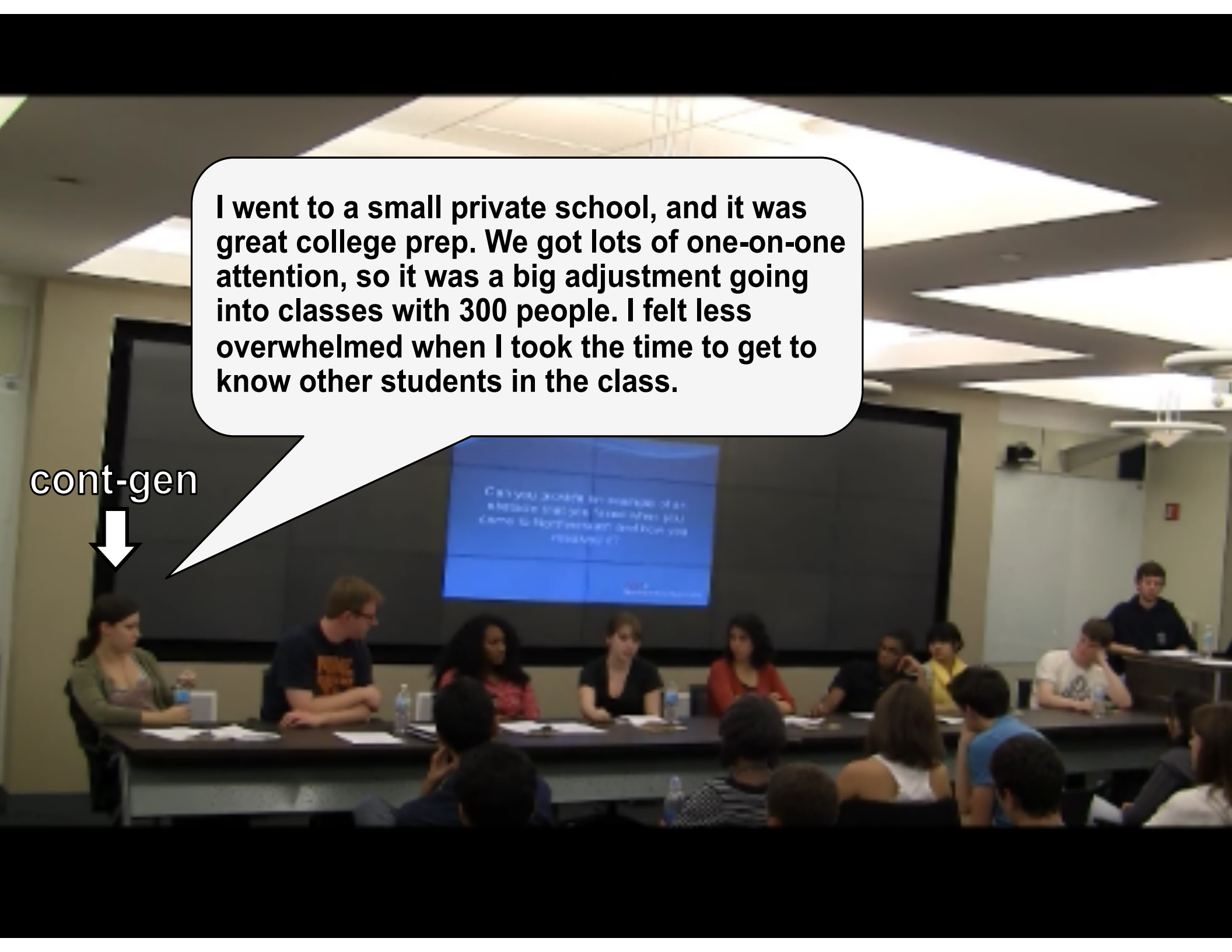


Because my parents didn't go to college, they weren't always able to provide me the advice I needed. So it was sometimes hard to figure out which classes to take and what I wanted to do in the future. But there are other people who can provide that advice, and I learned that I needed to rely on my adviser more than other students.



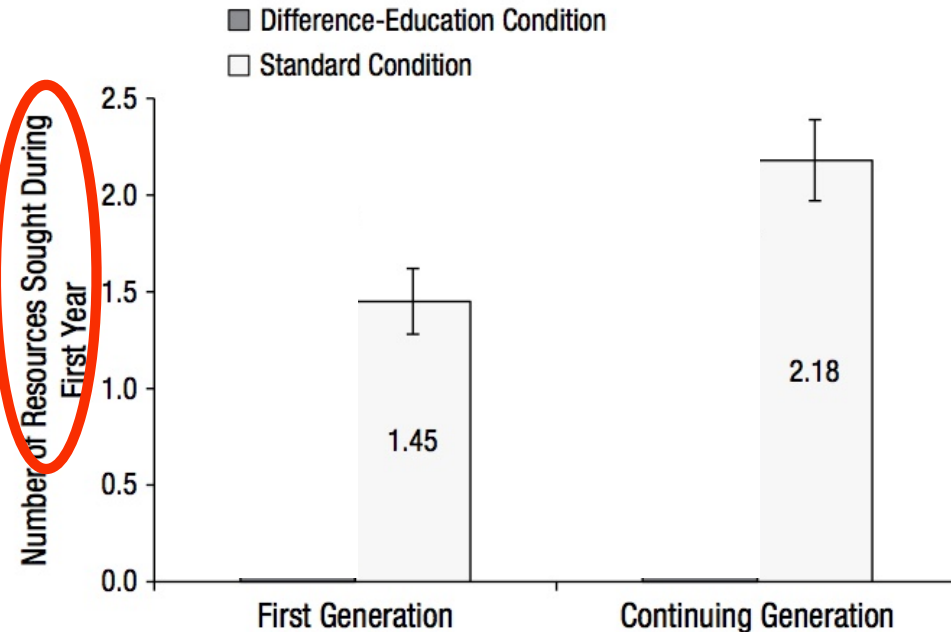
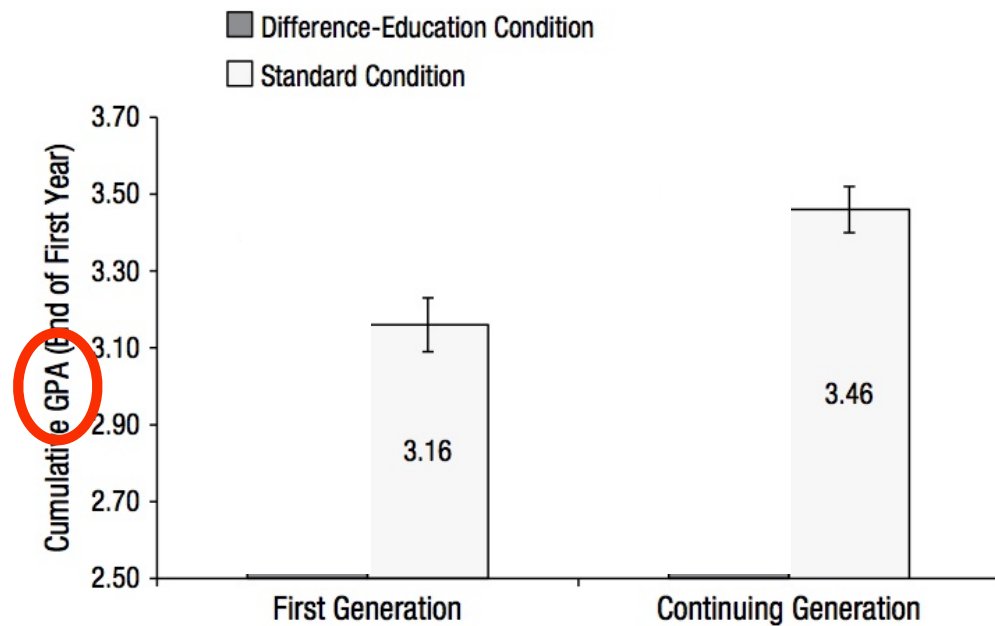
I went to a small private school, and it was great college prep. We got lots of one-on-one attention, so it was a big adjustment going into classes with 300 people. I felt less overwhelmed when I took the time to get to know other students in the class.

cont-gen



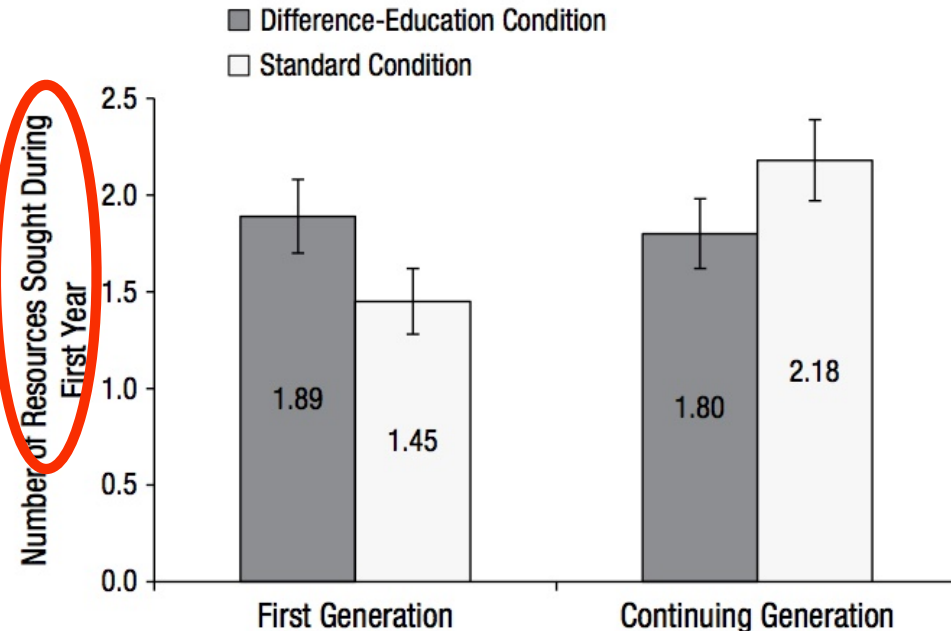
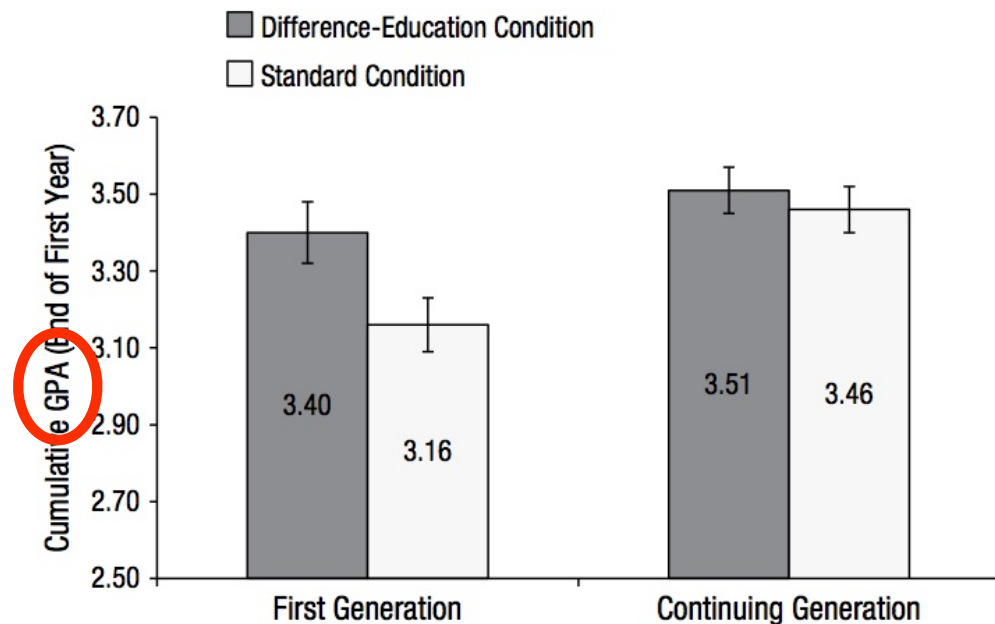
Closing the social-class achievement gap

- Two groups of participants: college students whose parents do / don't have 4-year degrees
- Two conditions: 'difference-education intervention' vs. control intervention



Closing the social-class achievement gap

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